

Enhanced Option Analytics Engine Documentation

This document explains the mathematics, pricing theory, implied volatility logic, CDF approximation, Greeks, theoretical option pricing, and architectural design of the single-header C++ option analytics engine. The engine supports: Black-Scholes model (Equity options) Black-76 model (Commodity/Futures options) Implied Volatility calculation Greeks computation Newton-Raphson convergence Low-latency inline execution

1. Variable Glossary

Variable	Meaning
S	Spot price of underlying asset
F	Futures price used in Black-76
K	Strike price
r	Risk-free interest rate
q	Dividend yield
T	Time to expiry in years
sigma	Volatility / Implied Volatility
d1	Probability-weighted intermediate Black variable
d2	Adjusted probability term
Theo Price	Model-derived fair value of the option
Market Price	Actual traded option premium
IV	Implied volatility extracted from market price

2. What is Theoretical Price?

Theoretical price (Theo Price) is the fair value of an option according to a pricing model. It is NOT the actual traded market price. Instead, it is the mathematically expected option value computed using: Underlying price Strike price Time to expiry Volatility Interest rate Dividend yield The implied volatility solver works by repeatedly changing volatility until: **Theoretical Price $\approx$ Market Price** This is the core idea behind implied volatility extraction.

3. Normal Distribution Approximation

The pricing models require the cumulative normal distribution function $N(x)$. Exact computation of $N(x)$ is computationally expensive because it involves integration. To improve performance, the engine uses the Abramowitz-Stegun polynomial approximation. This approximation is extremely popular in quantitative finance because: It is very fast Low memory overhead Good numerical stability High precision for trading systems

Approximation Formula

$$t = 1 / (1 + p \cdot x)$$

$$N(x) \approx 1 - [(((a5 \cdot t + a4) \cdot t + a3) \cdot t + a2) \cdot t + a1] \cdot t \cdot e^{(-x^2)}$$

Where:

$$\begin{aligned} a1 &= 0.254829592 \\ a2 &= -0.284496736 \\ a3 &= 1.421413741 \\ a4 &= -1.453152027 \\ a5 &= 1.061405429 \\ p &= 0.3275911 \end{aligned}$$

4. Black-Scholes Pricing Model

The Black-Scholes model prices European equity options. Core assumptions: Lognormal stock movement Constant volatility Continuous trading No arbitrage Constant interest rates

d1 Formula

$$d1 = [\ln(S/K) + (r - q + 0.5 \cdot \sigma^2) \cdot T] / (\sigma \cdot \sqrt{T})$$

d2 Formula

$$d2 = d1 - \sigma \cdot \sqrt{T}$$

Call Price Formula

$$\text{Call} = S \cdot e^{(-qT)} \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2)$$

Put Price Formula

$$\text{Put} = K \cdot e^{(-rT)} \cdot N(-d2) - S \cdot e^{(-qT)} \cdot N(-d1)$$

5. Black-76 Pricing Model

Black-76 is designed for options on futures contracts. Unlike Black-Scholes: It uses Futures price directly No dividend term exists Widely used in commodities and interest-rate derivatives

Black-76 Formula

$$\text{Call} = e^{(-rT)} \cdot [F \cdot N(d1) - K \cdot N(d2)]$$

6. Implied Volatility Solver

Implied volatility is the volatility value that makes the theoretical model price equal to the market option premium. The engine uses Newton-Raphson iteration: $\sigma_{\text{new}} = \sigma_{\text{old}} - (\text{TheoPrice} - \text{MarketPrice}) / \text{Vega}$ Why Newton-Raphson? Extremely fast convergence Industry-standard IV extraction technique Very low latency Suitable for HFT and RMS systems

7. Greeks and Risk Sensitivities

Greek	Meaning
Delta	Sensitivity to underlying movement

Gamma	Rate of change of Delta
Theta	Time decay
Vega	Sensitivity to volatility
Rho	Sensitivity to interest rates

Greeks are essential for: Risk Management Systems (RMS) Delta hedging Portfolio exposure analysis Volatility trading Scenario analysis

8. Engine Architecture

The engine is implemented as a header-only utility. Advantages: No linking overhead Easy integration Fully inline optimized Low-latency execution Portable across trading systems Execution Flow: Input market parameters Calculate Theo Price Calculate Vega Run Newton-Raphson loop Converge IV Compute Greeks Return analytics structure

9. Conclusion

This engine provides a compact but production-grade implementation of real-world option analytics. Suitable use-cases: Trading terminals RMS systems Algo trading engines Market simulators Quantitative research Because the utility is inline and header-only, it is ideal for ultra-low latency systems.